插图英语阅读小册

第4册 · 十篇故事

按推荐顺序阅读，保留原篇号。点击篇目即可跳转；每篇依次包含插图、英文原文、中文译文、词汇与回想。

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几个少年和一只狗发现的洞穴

The Cave Found by a Dog



谁发现了壁画，为什么今天只能看复制洞穴?

英文原文

Marcel Ravidat, 18, was on a woodland stroll near Montignac with Robot, his canine companion, when it began sniffing at a hollow left by a toppled tree. On 12 September 1940 he came back with three other boys for an adventurous excursion, and the four explorers climbed down into the darkness.

In later retellings, Ravidat embroidered his anecdote with a dramatic twist. Robot, he said, had fallen into the cave. That fanciful mishap is a fabrication. The animal only nosed about the hole, and the pivotal initiative was the boys' own. Ironically, the invented fall made the dog the hero of the story.

Some 680 painted figures and 1,500 engravings cover the rock. The artwork is usually dated to between 17,000 and 22,000 years ago; its creators, our distant ancestors, lived in antiquity. Horses, bulls, and stags are among the animals depicted in pigments on the murals. The paintings' fragility soon became clear.

Lascaux opened to the public on 14 July 1948, and tourism brought an influx of about 1,200 visitors a day. The throng exhaled carbon dioxide and added heat, humidity, and contaminants, and the deterioration of the paintings became plain to see. In 1963 the site closed to the public, followed by restoration and daily inspection.

Lascaux joined UNESCO's World Heritage List in 1979, and Lascaux II, a meticulously made partial facsimile, opened in 1983. A white fungus invaded the walls in a 2001 outbreak, and black stains spread after chemical treatment. Lascaux IV opened in December 2016. Admirers now meet the bulls only in a recreation, the originals being inaccessible to them.

中文译文

18岁的马塞尔·拉维达带着爱犬Robot，在蒙蒂尼亚克附近的林地散步时，Robot开始在一棵倒树留下的地洞旁嗅闻。1940年9月12日，拉维达带着另外三个男孩回来探险，四人爬进了黑暗中。

后来讲述这段经历时，拉维达添油加醋，加了一个戏剧性的转折：他说Robot掉进了洞穴。这个离奇的意外其实是编造的。狗只是在洞口嗅闻，作出关键探索决定的是少年们自己。讽刺的是，虚构的那一跤反而让狗成了故事里的英雄。

洞壁上布满约680个彩绘形象和1500幅雕刻图案。这些作品通常被断代为距今1.7万至2.2万年；创作者是生活在远古的先祖。壁画用颜料描绘了马、公牛、雄鹿等动物。然而，这些画作的脆弱很快显现出来。

拉斯科于1948年7月14日向公众开放，游客每天约有1200人。人群呼出二氧化碳，带来热量、湿气和污染物，壁画的退化变得清晰可见。1963年，原洞停止向公众开放，随后展开修复和每日检查。

1979年，拉斯科列入联合国教科文组织《世界遗产名录》。1983年，精心制作的局部复制洞穴“拉斯科二号”开放。2001年，一种白色真菌暴发并侵入洞壁；化学处理后，黑色污斑又蔓延开来。“拉斯科四号”于2016年12月开放。今天，参观者只能在复原品中欣赏那些公牛，原作所在的洞穴已不对他们开放。

核心词汇

先记这22项。连着下方的原文短语一起回想。

companion n.
伙伴
his canine companion

hollow n.
地面凹洞
a hollow left by a toppled tree

excursion n.
短途出游；探险
an adventurous excursion

embroider v.
添油加醋地讲述
embroidered his anecdote

anecdote n.
轶事；小故事
embroidered his anecdote

fabrication n.
编造；虚构
That fanciful mishap is a fabrication

pivotal adj.
关键的
the pivotal initiative

initiative n.
主动行动；主动性
the pivotal initiative

painted figure 名词短语
彩绘形象
680 painted figures

engraving n.
雕刻图案
1,500 engravings cover the rock

ancestor n.
祖先
our distant ancestors

depict v.
描绘
the animals depicted in pigments

pigment n.
颜料
depicted in pigments on the murals

mural n.
壁画
depicted in pigments on the murals

fragility n.
脆弱性
The paintings' fragility

influx n.
大量涌入
an influx of about 1,200 visitors

contaminant n.
污染物
heat, humidity, and contaminants

deterioration n.
退化；恶化
the deterioration of the paintings

restoration n.
修复
followed by restoration

facsimile n.
复制品
a meticulously made partial facsimile

recreation n.
复原品；再现
only in a recreation

inaccessible adj.
无法进入或接近的
the originals being inaccessible

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

woodland n.
林地

stroll n.
散步

canine adj.
犬类的

sniff v.
嗅闻

topple v.
倒下；倾倒

adventurous adj.
富于冒险的

explorer n.
探险者

darkness n.
黑暗

retelling n.
重新讲述

dramatic adj.
戏剧性的

fall into 动词短语
掉进

fanciful adj.
想象出来的；不真实的

mishap n.
意外；小事故

ironically adv.
讽刺的是

invent v.
编造

cover v.
布满；覆盖

artwork n.
艺术作品

creator n.
创作者

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

antiquity n.
远古；古代

bull n.
公牛

open to the public 动词短语
向公众开放

tourism n.
旅游；旅游业

throng n.
密集的人群

exhale v.
呼出

carbon dioxide 名词短语
二氧化碳

humidity n.
湿度；湿气

plain to see 形容词短语
显而易见的

inspection n.
检查

meticulously adv.
精细地；一丝不苟地

partial adj.
局部的；部分的

fungus n.
真菌

invade v.
侵入

outbreak n.
暴发

stain n.
污斑

chemical treatment 名词短语
化学处理

admirer n.
欣赏者；仰慕者

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 That fanciful mishap is a _____.

02 the pivotal _____ was the boys' own.

03 The paintings' _____ soon became clear.

04 the originals being _____ to them.

文章主线

- 发现 1940年，狗在地洞旁嗅闻引起拉维达注意；他随后与另外三名少年下洞，发现拉斯科史前壁画。
- 澄清 后来拉维达把故事说成狗掉进洞里；其实狗只在洞口嗅闻，下洞探险是少年们主动决定的。
- 瑰宝 洞内约有680个彩绘形象和1500幅雕刻图案，常被断代为距今约1.7万至2.2万年，描绘马、牛、鹿等动物。
- 受损 1948年开放后，游客带来二氧化碳、热量、湿气和污染，壁画退化，1963年闭洞。
- 保护 闭洞后仍出现真菌、黑斑；1983年和2016年先后开放复制洞穴，公众如今只能看复制品。

参考答案　1 fabrication 2 initiative 3 fragility 4 inaccessible

黄石公园狼影再现

Shadows of Wolves Over Yellowstone



狼回来了，河谷恢复就都该归功于狼吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Yellowstone's iconic grey wolves had been exterminated by 1926. On 12 January 1995 eight wolves captured in Alberta, Canada, were hauled to the park in crates; six more followed. Released from enclosures in late March, they roamed the wilderness.

In February 2014 a viral video narrated by writer George Monbiot spun a compelling narrative: elk, thinned by wolves and wary of them, retreated from valleys, letting willows flourish and riverbanks stabilise. Ecologists call this a trophic cascade: predators eat or deter plant-eaters, and vegetation rebounds.

Yet the census undermined the thinning claim: elk on the park's northern range peaked at about 19,000 in January 1994 and were dwindling before wolves arrived, depleted by hunters, drought, and bears. Fear looked weak too: in June 2013 Arthur Middleton's University of Wyoming team found that collared elk east of the park kept grazing unless wolves ventured within half a mile.

Trees fared no better: in November 2021 Utah State University's Elaine Brice showed that sampling only the tallest aspens overstated regrowth four- to sevenfold. The deeper flaw was water. Colorado State University's Tom Hobbs and David Cooper had excluded elk with fences and built artificial beaver dams. In January 2024 they reported: eroding streams had cut so deep that stunted willows no longer reached water.

In January 2025 Oregon State University's William Ripple, an advocate of the hypothesis, reported one of the world's strongest cascades; critics disputed his methods. The question is unresolved, but the wolves' shadow looks fainter than the fable suggested, and the science messier. A discipline willing to revise its cherished example may deserve more trust than the fable ever did.

中文译文

到1926年，黄石公园标志性的灰狼已被消灭。1995年1月12日，八只在加拿大艾伯塔省捕获的狼被装进板条箱运到公园，随后又来了六只。3月下旬，它们离开围栏，开始在荒野中活动。

2014年2月，一段由作家乔治·蒙比奥特解说的热门视频讲了一个动人的故事：狼捕杀北美马鹿，让鹿群数量减少，又因使它们害怕而迫使其离开河谷；柳树于是繁盛起来，河岸也更稳固。生态学家把这类连锁影响称为“营养级联”：捕食者吃掉或吓阻食草动物，继而让植被恢复。

然而，普查数据削弱了鹿群因狼而减少的说法。公园北部草场的北美马鹿在1994年1月达到约1.9万头的高峰，狼还没来，鹿群就已因猎捕、干旱和熊而减少。恐惧导致行为改变的证据也不强：2013年6月，阿瑟·米德尔顿领导的怀俄明大学团队发现，公园以东戴追踪项圈的鹿仍照常吃草，除非狼进入半英里以内。

树木方面的证据也不那么理想。2021年11月，犹他州立大学的伊莱恩·布赖斯指出，只抽样测量最高的山杨，会把再生程度夸大四到七倍。更深层的问题在水上。科罗拉多州立大学的汤姆·霍布斯和戴维·库珀用围栏排除鹿群，并修建人工海狸坝。2024年1月，他们报告说，溪流侵蚀使河床下切得太深，生长受阻的柳树已无法从根部够到水。

2025年1月，支持营养级联假说的俄勒冈州立大学学者威廉·里普尔发表研究，称黄石的级联效应是世界上最强的案例之一；批评者则质疑其方法。问题仍未解决，但狼对植被恢复的影响似乎不像那个简单故事所讲的那么强，科学证据也更复杂。一门愿意修正自己珍视案例的学科，也许比那个寓言般的故事更值得信任。

本文的elk指北美马鹿，不是中国物种名“麋鹿”所指的动物。争论的是狼促成植被恢复的程度，不是狼是否具有任何生态作用。

核心词汇

先记这22项。连着下方的原文短语一起回想。

iconic adj.
标志性的
Yellowstone's iconic grey wolves

exterminate v.
彻底消灭
had been exterminated by 1926

enclosure n.
围栏区域
Released from enclosures

wilderness n.
荒野
they roamed the wilderness

compelling adj.
引人入胜的
a compelling narrative

narrative n.
故事；叙事
a compelling narrative

wary of 形容词短语
提防；警惕
wary of them

trophic cascade n.
营养级联
a trophic cascade

predator n.
捕食者
predators eat or deter plant-eaters

deter v.
吓阻
deter plant-eaters

vegetation n.
植被
vegetation rebounds

census n.
普查
the census

undermine v.
削弱；动摇
undermined the thinning claim

dwindle v.
逐渐减少
dwindling before wolves arrived

deplete v.
使减少；消耗
depleted by hunters

sample v.
抽样测量
sampling only the tallest aspens

overstate v.
夸大
overstated regrowth four- to sevenfold

regrowth n.
再生
overstated regrowth four- to sevenfold

hypothesis n.
假说
an advocate of the hypothesis

dispute v.
质疑；对……提出异议
critics disputed his methods

unresolved adj.
尚未解决的
The question is unresolved

revise v.
修正
revise its cherished example

文中词汇与搭配

补充词汇共32项。保留每个表达在本文中的意思。

capture v.
捕获

haul v.
拖运

crate n.
板条箱

roam v.
漫游；四处活动

viral video n.
迅速走红的视频

be narrated by 动词短语
由……解说

spin v.
编织（故事）

retreat from 动词短语
从……撤离

willow n.
柳树

flourish v.
繁盛

stabilise v.
变得稳固；稳定

rebound v.
恢复；回升

peak at 动词短语
达到……峰值

hunter n.
猎人

drought n.
干旱

collared adj.
戴有追踪项圈的

graze v.
吃草

venture v.
冒险进入

fare no better 动词短语
情况也没有更好

aspen n.
山杨

flaw n.
问题；缺陷

exclude v.
排除；挡在外面

artificial adj.
人工的

dam n.
坝

erode v.
侵蚀

stunted adj.
生长受阻的

advocate n.
倡导者

faint adj.
较弱的；不明显的

fable n.
寓言式的故事

messy adj.
更错综复杂的

discipline n.
学科

cherished adj.
珍视的

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a viral video narrated by writer George Monbiot spun a _____ narrative

02 elk on the park's northern range peaked at about 19,000 in January 1994 and were _____ before wolves arrived, depleted by hunters, drought, and bears.

03 Trees fared no better: in November 2021 Utah State University's Elaine Brice showed that sampling only the tallest aspens _____ regrowth four- to sevenfold.

04 In January 2025 Oregon State University's William Ripple, an advocate of the _____, reported one of the world's strongest cascades; critics disputed his methods.

文章主线

- 回归 黄石公园的灰狼到1926年已被消灭，1995年重新引入；流行视频称狼捕杀并吓退北美马鹿，让柳树复苏、河岸稳固。
- 质疑 这类捕食者经食草动物影响植被的连锁效应，叫“营养级联”；但鹿群在狼来前已因猎捕、干旱和熊而减少。
- 证据 除非狼接近至半英里内，公园以东被追踪的北美马鹿仍照常吃草；只测最高山杨，还会把再生夸大四至七倍。
- 瓶颈 围栏与人工海狸坝实验揭示，河床下切让部分柳树够不到水；减少北美马鹿啃食本身，不足以保证柳树恢复。
- 争议 2025年支持者仍称级联很强，批评者则质疑方法；狼促成植被恢复的程度仍有争议，应随证据修正这个经典案例。

参考答案　1 compelling 2 dwindling 3 overstated 4 hypothesis

走出苏联的俄罗斯方块

The Game That Escaped the Soviet Union



游戏风靡世界后，为什么作者多年未分到游戏收益？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In June 1984 Alexey Pajitnov, a 29-year-old Moscow coder, grew engrossed in a diversion he had built. His primitive Elektronika 60 had a purely textual display, so every geometric piece of four tiles was assembled from characters. The result proved addictive. He called it Tetris, a fusion of the Greek tetra- and tennis, his favourite pastime.

In the Soviet Union, the state, not its salaried designer, was the proprietor of this intellectual property, and Pajitnov held no entitlement to it. Copies mushroomed freely on floppy disks across the Eastern bloc.

In 1986 Robert Stein, an enterprising British software vendor, stumbled upon Tetris in Hungary. In his ignorance of Elorg, the state bureau regulating Soviet software exports, he hastily hawked rights he did not yet legitimately hold. The rights were shuffled onward through a chaotic muddle of overlapping deals among Western firms, one owned by the magnate Robert Maxwell.

In February 1989 Henk Rogers, a resourceful Dutch-born publisher, reached Moscow on a tourist visa and found his way to Elorg. Stein and Kevin Maxwell, the magnate's son, were in Moscow that same hectic week of negotiations, each a bidder. Pajitnov felt a fondness for Rogers, and Nintendo scooped the handheld rights, then worldwide console rights.

Nintendo bundled Tetris with its pocket-sized Game Boy, and that version shipped over 35 million copies. Pajitnov reaped none of the spoils for years. His earnings began only after his arrangement's expiry around 1995. In 1996 he and Rogers founded The Tetris Company to manage the rights.

The man whose puzzle once spread for nothing later helped decide who could sell it.

中文译文

1984年6月，29岁的莫斯科程序员阿列克谢·帕基特诺夫，沉迷于自己编写的一款小游戏。他那台简陋的Elektronika 60电脑只能显示文字，于是每个由四个小格构成的几何块，都用字符拼成。游戏让人欲罢不能。他把希腊语前缀tetra-与自己最爱的消遣——网球的英文tennis——结合起来，给它取名Tetris，也就是俄罗斯方块。

在苏联，这项知识产权归国家所有，并不属于领取工资的设计者，帕基特诺夫没有支配它的权利。游戏复制品通过软盘，在东欧集团各地自由传播、迅速增多。

1986年，富有进取心的英国软件商罗伯特·斯坦因在匈牙利偶然发现俄罗斯方块。他不知道Elorg是监管苏联软件出口的国家机构，尚未合法取得权利，便草率地兜售版权。权利在西方公司之间不断转手，各种相互重叠的交易混成一团；其中一家公司属于大亨罗伯特·马克斯韦尔。

1989年2月，出生于荷兰、足智多谋的发行商亨克·罗杰斯持旅游签证来到莫斯科，设法找到Elorg。同一个忙乱的谈判周里，斯坦因与大亨之子凯文·马克斯韦尔也在莫斯科，分别参与竞标。帕基特诺夫对罗杰斯颇有好感。任天堂最终抢得掌机版权，随后又拿到全球游戏主机版权。

任天堂把俄罗斯方块与口袋大小的Game Boy捆绑销售，这个版本出货超过3500万份。帕基特诺夫多年未能分享游戏带来的商业收益；直到原有协议于1995年前后到期，才开始从游戏中获得收益。1996年，他与罗杰斯创办俄罗斯方块公司，管理相关权利。

当年，他的益智游戏曾免费流传；后来，他也参与决定谁可以出售它。

核心词汇

先记这22项。连着下方的原文短语一起回想。

engrossed in 形容词短语
全神贯注于；沉迷于
grew engrossed in a diversion

textual adj.
文字的；文本的
a purely textual display

be assembled from 动词短语
由……拼成
assembled from characters

fusion n.
融合
a fusion of the Greek tetra-

proprietor n.
所有者
the proprietor

intellectual property 名词短语
知识产权
this intellectual property

entitlement n.
拥有某物的权利
held no entitlement to it

mushroom v.
迅速增多
Copies mushroomed freely

enterprising adj.
富有进取心的
enterprising British software vendor

vendor n.
销售商
enterprising British software vendor

stumble upon 动词短语
偶然发现
stumbled upon Tetris in Hungary

regulate v.
监管
regulating Soviet software exports

legitimately adv.
合法地
did not yet legitimately hold

overlap v.（分词作定语）
相互重叠
overlapping deals among Western firms

resourceful adj.
足智多谋的
a resourceful Dutch-born publisher

negotiation n.
谈判
hectic week of negotiations

bidder n.
竞标者
each a bidder

bundle v.
捆绑销售
Nintendo bundled Tetris

reap v.
收获；获得
reaped none of the spoils

earnings n.
收入；此处指游戏收益
His earnings began only after

arrangement n.
协议；安排
his arrangement's expiry

expiry n.
到期
his arrangement's expiry around 1995

文中词汇与搭配

补充词汇共43项。保留每个表达在本文中的意思。

Moscow 专名
莫斯科

coder n.
程序员

diversion n.
消遣；游戏

primitive adj.
简陋的

purely adv.
完全地；纯粹地

geometric adj.
几何的

tile n.
小格；方块

character n.
字符

addictive adj.
让人上瘾的

Tetris 专名
俄罗斯方块

Greek adj.
希腊语的

pastime n.
消遣

salaried adj.
领取工资的

designer n.
设计者

freely adv.
自由地；不受限制地

floppy disk 名词短语
软盘

Eastern bloc 专名
东欧集团

software n.
软件

Hungary 专名
匈牙利

in prep.
处于；与ignorance of连用

ignorance n.
不知情；缺乏了解

of prep.
对；引出不了解的对象

文中词汇与搭配 续

补充词汇共43项。保留每个表达在本文中的意思。

state bureau 名词短语
国家机构

hastily adv.
草率地；仓促地

hawk v.
兜售

shuffle v.
不断倒手；来回转移

onward adv.
继续向前；此处指继续转手

chaotic adj.
混乱的

muddle n.
混乱局面

magnate n.
商业大亨

publisher n.
发行商

visa n.
签证

find one's way to 动词短语
设法找到；来到

hectic adj.
忙乱的

fondness n.
好感；喜爱

Nintendo 专名
任天堂

scoop v.
抢先获得

console n.
游戏主机

pocket-sized adj.
口袋大小的

version n.
版本

spoils n.
获利；收益

found v.
创办

puzzle n.
益智游戏

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the state, not its salaried designer, was the proprietor of this _____

02 Copies _____ freely on floppy disks across the Eastern bloc.

03 Nintendo _____ Tetris with its pocket-sized Game Boy

04 His earnings began only after his arrangement's _____ around 1995.

文章主线

- 诞生 1984年，苏联程序员帕基特诺夫在只能显示文字的电脑上，用字符做出了俄罗斯方块。
- 归属 游戏版权属于苏联国家，作者没有支配权，复制品则通过软盘在东欧集团迅速流传。
- 混战 英国软件商斯坦因不知Elorg掌管苏联软件出口，未获授权就转卖版权，造成西方公司权利重叠。
- 突破 1989年，发行商罗杰斯找到Elorg谈判，任天堂获掌机及全球主机版权，Game Boy版出货3500多万份。
- 获益 帕基特诺夫到约1995年旧协议到期才开始获得游戏收益；1996年与罗杰斯成立公司，参与管理版权。

参考答案　1 intellectual property 2 mushroomed 3 bundled 4 expiry

多莉：一只羊的复制

Dolly, the Copied Sheep



白脸小羊为何由黑脸母羊生下?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Dolly was born at Roslin, Scotland, on 5 July 1996. Her genome came from a Finn Dorset ewe's mammary cell, fused with a Blackface egg stripped of its nucleus. A Blackface surrogate carried the embryo through pregnancy, yet her white face was proof of a genetically different lineage. She alone survived from 277 attempts, an extraordinary failure rate.

Dolly is popularly called the first clone, a misconception. Her predecessors included frogs cloned from transplanted nuclei in 1962 and Roslin's embryo-derived duo, Megan and Morag, in 1995. She was genuinely the first mammal cloned from an adult cell. Roslin's announcement on 22 February 1997 coincided with the paper, and her name, a mammary pun, honoured Dolly Parton.

The news provoked uproar and apprehension about the ethics of human cloning. In March 1997 Clinton prohibited federal funding for such experiments. In 1999 her telomeres, or chromosome tips, were reported shortened by about 20% for her age. That fed anxiety; some speculated she was ageing prematurely.

Dolly was put down, aged six, on 14 February 2003, with arthritis and a progressive lung disease caused by a sheep virus. Indoor confinement probably contributed. In July 2016, 13 clones, four from her cell line, were ageing normally, without hypertension or diabetes. In November 2017, her skeletal X-rays showed ordinary arthritis, and fears of early onset were judged unfounded.

She bore six lambs by natural mating; Bonnie, first of these descendants, arrived in April 1998. Since 2003 her stuffed body, now famously an icon and memorial, has been encased in the National Museum of Scotland. Behind glass, her white face plainly testifies to her Finn Dorset ancestry.

中文译文

多莉于1996年7月5日在苏格兰罗斯林出生。她的基因组来自一只芬兰多塞特母羊的乳腺细胞；研究人员把这个细胞与一枚已去除细胞核的黑脸羊卵细胞融合。胚胎由一只黑脸羊代孕并孕育至出生，但多莉的白脸显示，她的遗传血统与代孕母羊不同。277次尝试中，只有她存活下来，失败率高得惊人。

人们常把多莉称为第一个克隆体，这是误解。在她之前，已有1962年通过移植细胞核克隆的青蛙，以及罗斯林1995年用胚胎细胞克隆出的梅根和莫拉格两只羊。她真正创下的纪录，是成为首只由成年细胞克隆的哺乳动物。罗斯林于1997年2月22日公布消息，相关论文也同时发表。她的名字带有乳腺相关的双关意味，向多莉·帕顿致意。

消息引起轩然大波，也引发人们对克隆人伦理的忧虑。1997年3月，克林顿禁止以联邦资金资助这类实验。1999年有报告称，多莉的端粒，即染色体末端，比她这个年龄应有的长度短约20%。这加重了焦虑，有人推测她正在过早衰老。

2003年2月14日，六岁的多莉被实施安乐死。她当时患有关节炎，以及由一种绵羊病毒引起、不断恶化的肺病；长期圈养在室内可能也是促成病情的因素。2016年7月，一项研究中的13只克隆羊都在正常衰老，没有高血压或糖尿病，其中4只来自与多莉相同的细胞系。2017年11月，对多莉骨骼X光片的检查显示，她患的是普通关节炎，对过早发病的担忧被认为缺乏根据。

多莉通过自然交配生下六只小羊。第一个后代邦妮于1998年4月出生。自2003年起，她的填充标本一直陈列在苏格兰国家博物馆的玻璃展柜中，如今已是广为人知的象征和纪念物。隔着玻璃，她的白脸依然清楚地表明其芬兰多塞特祖源。

“基因组来自乳腺细胞”主要指细胞核中的遗传物质；去核卵细胞仍保留自身的线粒体遗传物质，代孕母羊则负责孕育胚胎。

核心词汇

先记这22项。连着下方的原文短语一起回想。

genome n.
基因组
Her genome came

fuse with 动词短语
与……融合
fused with a Blackface egg

strip of 动词短语
去除；剥去
stripped of its nucleus

nucleus n.
细胞核
stripped of its nucleus

surrogate n.
代孕母羊；代孕者
Blackface surrogate

embryo n.
胚胎
carried the embryo through pregnancy

lineage n.
血统；世系
a genetically different lineage

clone n.
克隆体
the first clone

misconception n.
误解；错误观念
a misconception

mammal n.
哺乳动物
the first mammal cloned

coincide with 动词短语
与……同时发生
coincided with the paper

provoke v.
引起；激起
The news provoked uproar

apprehension n.
忧虑；担心
apprehension about the ethics

ethics n.
伦理；道德规范
the ethics of human cloning

prohibit v.
禁止
Clinton prohibited federal funding

speculate v.
推测
some speculated

prematurely adv.
过早地
she was ageing prematurely

progressive adj.
病情逐渐加重的
a progressive lung disease

confinement n.
限制活动；圈养
Indoor confinement

onset n.
发病；开始出现
fears of early onset

unfounded adj.
没有根据的
were judged unfounded

ancestry n.
祖源；世系
her Finn Dorset ancestry

文中词汇与搭配

补充词汇共34项。保留每个表达在本文中的意思。

Scotland 专有名词
苏格兰

pregnancy n.
孕期；怀孕

proof of 名词短语
……的证据

genetically adv.
从遗传上说

extraordinary adj.
异乎寻常的

popularly adv.
通常；普遍地

predecessor n.
先前的同类；前身

transplant v.
移植

duo n.
一对；两个一组

genuinely adv.
真正地；确实

announcement n.
公告；宣布

pun n.
双关语

honour v.
向……致意

uproar n.
轩然大波

chromosome n.
染色体

shorten v.
缩短

anxiety n.
焦虑

put down 动词短语
对动物实施安乐死

arthritis n.
关节炎

lung n.
肺

virus n.
病毒

contribute v.
是促成因素之一

hypertension n.
高血压

diabetes n.
糖尿病

skeletal adj.
骨骼的

mating n.
交配

descendant n.
后代

stuffed adj.
制成填充标本

famously adv.
众所周知地

icon n.
象征；标志性事物

memorial n.
纪念物

encase in 动词短语
封装在……里面

plainly adv.
清楚地

testify to 动词短语
表明；证明

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Dolly is popularly called the first clone, a _____.

02 Her _____ came from a Finn Dorset ewe's mammary cell

03 some speculated she was ageing _____.

04 fears of early onset were judged _____.

文章主线

- 诞生 研究者将去掉细胞核的卵细胞与成年母羊乳腺细胞融合，培育胚胎，1996年由代孕母羊产下多莉。
- 突破 277次尝试仅多莉存活；她是首只由成年细胞克隆的哺乳动物，此前已能用胚胎细胞克隆羊。
- 争议 成果引发人类克隆伦理争论；多莉的端粒（染色体末端）比同龄羊短约20%，又引起早衰猜测。
- 离世 多莉自然交配育下六只小羊；患有关节炎和病毒性肺病，六岁时被实施安乐死，不能据此认定克隆导致早衰。
- 复核 后续研究中13只克隆羊正常衰老；多莉骨骼复查也未支持关节炎过早出现的担忧。

参考答案　1 misconception 2 genome 3 prematurely 4 unfounded

不合拍的风：塔科马大桥的陨落

The Wind That Kept No Time: The Fall of Galloping Gertie



风没按节拍吹，大桥为何还是倒了？

READ 纯净阅读 | CHECK 中文核对 | LEARN 词汇记忆 | RECALL 回想巩固

英文原文

On the morning of 7 November 1940, a sustained wind of about 40 miles an hour set the Tacoma Narrows Bridge twisting. Its deck tilted from side to side in ever more violent oscillations until most of the main span tore apart and tumbled into the water. No one died. Footage of the spectacle would later fascinate students worldwide.

The bridge had opened in July 1940 as the third-longest suspension span in the world at the time, its roadway hung from steel cables. Girders only eight feet deep stiffened the deck, making it slender, elegant, and economical, but far from rigid. Even faint breezes made it heave, earning the nickname Galloping Gertie.

Textbooks have long invoked resonance: a periodic push that matches an object's natural frequency can amplify its motion, as when well-timed shoves propel a child on a swing. The analogy is intuitive but deceptive. That morning's wind did not pulse in time with the bridge; it blew fairly steadily.

The mechanism was subtler. Each twist altered the passing airflow, which reinforced the next twist: a feedback loop that kept tapping the wind for energy. In March 1941 a federal panel blamed the deck's flexibility and wing-like lift. Later analysis called the mechanism flutter; in February 1991 two engineers asserted in a physics journal that textbooks had erred.

The panel urged wind-tunnel tests, and trials on scaled-down replicas gradually became customary for suspension bridges. A sturdier replacement opened in October 1950. Gertie's fall may still instruct engineers, if read accurately: the wind never kept time with the bridge, the bridge's own motion turned a sustained wind destructive.

中文译文

1940年11月7日上午，时速约40英里的持续大风使塔科马海峡大桥开始扭转。桥面左右倾斜，振荡越来越剧烈，最终主跨的大部分断裂，坠入水中。无人遇难。这场景象的影像，后来吸引了世界各地的学生。

大桥于1940年7月通车，其悬索桥跨径当时位居世界第三，路面由钢缆悬吊。桥面只靠高八英尺的钢梁加强，因此显得纤细、优美又经济，却远不够刚硬。即使微风也会让它起伏，因而得了“奔腾的格蒂”这个绰号。

教科书长期用共振来解释这起事故：如果周期性的推力与物体的固有频率相合，就能放大运动，就像踩准节拍推秋千上的孩子。这个类比直观，却会误导人。那天上午的风并没有跟着大桥的节拍一阵阵吹来，而是相当平稳。

实际机制更微妙。桥面每一次扭转都会改变掠过的气流，气流又加强下一次扭转，形成不断从风中获取能量的反馈回路。1941年3月，一个联邦调查委员会把事故归因于桥面的柔性和类似机翼的升力。后来的分析把这一机制称为颤振；1991年2月，两位工程师在一份物理学期刊上指出，教科书的解释错了。

委员会力主做风洞试验，此后用缩尺模型测试悬索桥逐渐成为惯例。更坚固的新桥于1950年10月通车。只要读懂真正的机制，格蒂的倒塌仍能给工程师以启示：风从未与桥合拍，是桥自身的运动，让一股持续的风变得具有破坏力。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sustained adj.
持续的
a sustained wind of about 40 miles an

oscillation n.
振荡
side in ever more violent oscillations

tear apart phr.v.
撕裂；断裂
the main span tore apart and tumbled

footage n.
影片片段；影像
Footage of the spectacle would later

suspension n.
悬吊；悬索结构
as the third-longest suspension span

span n.
桥跨；跨径
as the third-longest suspension span

rigid adj.
刚硬的；不易变形的
and economical, but far from rigid.

heave v.
上下起伏
Even faint breezes made it heave,

invoke v.
援引；用作解释
Textbooks have long invoked resonance:

resonance n.
共振
Textbooks have long invoked resonance:

periodic adj.
周期性的
periodic push that matches an object's

frequency n.
频率
object's natural frequency can amplify

amplify v.
放大；增强
object's natural frequency can amplify

analogy n.
类比
The analogy is intuitive but

intuitive adj.
直观易懂的
analogy is intuitive but deceptive.

deceptive adj.
具有误导性的
analogy is intuitive but deceptive.

mechanism n.
机制；机理
The mechanism was subtler.

feedback n.
反馈
a feedback loop that kept tapping the

flexibility n.
柔性；易弯曲性
panel blamed the deck's flexibility

scaled-down adj.
按比例缩小的
scaled-down replicas gradually became

customary adj.
惯常的
replicas gradually became customary

destructive adj.
具有破坏力的
turned a sustained wind destructive.

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

twist v.
扭转

deck n.
桥面

tilt v.
倾斜

violent adj.
剧烈的

tumble v.
翻落；坠落

spectacle n.
壮观的景象

fascinate v.
使着迷；吸引

hang from phr.v.
悬挂于

girder n.
大梁；钢梁

stiffen v.
使更坚硬；加固

slender adj.
纤细的

elegant adj.
优美的

economical adj.
经济实惠的

faint adj.
微弱的

breeze n.
微风

nickname n.
绰号

textbook n.
教科书

well-timed adj.
时机恰当的

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

shove n.
推；猛推

propel v.
推动

pulse v.
有节奏地起伏

steadily adv.
平稳持续地

subtle adj.
更微妙的

alter v.
改变

reinforce v.
加强

tap v.
汲取；利用

panel n.
专家小组；调查委员会

assert v.
断言；明确提出

err v.
犯错

wind-tunnel adj.
风洞的

replica n.
复制模型

gradually adv.
逐渐地

sturdy adj.
更坚固的

replacement n.
替代物；此处指新桥

instruct v.
指导；给以启示

accurately adv.
准确地

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a _____ wind of about 40 miles an hour

02 can _____ its motion,

03 a _____ loop that kept tapping the wind for energy.

04 gradually became _____ for suspension bridges.

文章主线

- 事故 1940年，刚通车数月的塔科马大桥在持续大风中剧烈扭转、断裂坍塌，无人遇难。
- 隐患 轻巧经济的桥面刚度不足，连微风都能使其起伏，早已有“奔腾的格蒂”之名。
- 纠误 课本常用共振解释事故，但当天的风相当平稳，并没有按桥的节拍反复推它。
- 机制 桥的扭转改变气流，气流又加强下一次扭转，不断从风中取能；这一机制称为颤振。
- 后续 事故推动悬索桥模型风洞试验成为惯例；更坚固的新桥于1950年通车。

参考答案　1 sustained 2 amplify 3 feedback 4 customary

改变世界的科学巨匠

The Scientific Giant Who Changed the World



爱因斯坦因为什么获得诺贝尔奖?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On 29 May 1919, as a total eclipse obscured the sun, two British expeditions photographed stars in its apparent vicinity. Arthur Eddington observed from Príncipe, an island off West Africa; Crommelin and Davidson from Sobral, in north-eastern Brazil. They were testing general relativity's bold prediction that the sun's gravity would deflect starlight.

On 6 November, at a joint meeting of the Royal Society and the Royal Astronomical Society, the findings were hailed as confirming it. The Times proclaimed a "Revolution in Science"; a New York Times headline read "Lights All Askew in the Heavens". Almost overnight, Einstein, eminent among physicists, became a global celebrity.

Stockholm proved harder to dazzle. The Nobel physics committee valued rigorous experimental confirmation, and committee member Allvar Gullstrand, a Nobel-winning eye specialist, deemed the empirical support for relativity insufficient. Since no nomination fulfilled Nobel's criteria, the 1921 prize was reserved for a year, as the statutes permit.

On 9 November 1922 the Royal Swedish Academy of Sciences conferred it on Einstein, citing his services to theoretical physics, especially his discovery of the law of the photoelectric effect. That law, which governs how light ejects electrons from metal, helped underpin quantum theory. A letter the next day clarified that the prize took no account of relativity. Einstein, then aboard a steamer bound for Japan, would miss the December ceremony.

On 11 July 1923, in Gothenburg, Sweden, he delivered his Nobel lecture before a congress of Nordic scientists, devoting it to the fundamental ideas and problems of relativity. The theory the committee looked past reached a Nobel podium after all, carried there by the man the prize had honoured.

中文译文

1919年5月29日，日全食遮住太阳时，两支英国考察队拍摄了天空中看似靠近太阳的恒星。阿瑟·爱丁顿在西非外海的普林西比岛观测，克罗姆林和戴维森则在巴西东北部的索布拉尔观测。他们要检验广义相对论的一个大胆预言：太阳的引力会让星光发生偏折。

同年11月6日，在英国皇家学会和皇家天文学会的联席会议上，这些观测结果被盛赞为证实了该预言。《泰晤士报》宣告“科学革命”；《纽约时报》则以“天上的光全歪了”为标题。几乎一夜之间，原本已在物理学界享有盛名的爱因斯坦，变成了全球名人。

斯德哥尔摩的诺奖评审却没那么容易被打动。诺贝尔物理学委员会重视严格的实验证实，而委员阿尔瓦·古尔斯特兰德——一位获得过诺奖的眼科专家——认为相对论的实证支持不足。由于没有一项提名符合诺贝尔规定的标准，1921年的物理学奖按照章程保留一年。

1922年11月9日，瑞典皇家科学院把这个奖授予爱因斯坦，表彰他对理论物理学的贡献，尤其是发现光电效应定律。这一定律揭示光如何使金属释放电子，并帮助奠定了量子理论的基础。授奖决定次日的一封信明确说明，奖项并未考虑相对论。爱因斯坦当时乘轮船前往日本，因此未能参加12月的颁奖典礼。

1923年7月11日，爱因斯坦在瑞典哥德堡的北欧科学家大会上发表诺贝尔演讲，专讲相对论的基本思想与问题。那个被委员会绕开的理论，终究登上了诺贝尔讲台；把它带上去的，正是获得这个奖的人。

核心词汇

先记这22项。连着下方的原文短语一起回想。

eclipse n.
日食；月食
as a total eclipse obscured the sun,

obscure v.
遮住；使模糊
as a total eclipse obscured the sun,

expedition n.
考察队；远征
British expeditions photographed stars

vicinity n.
附近
stars in its apparent vicinity.

deflect v.
使偏折
sun's gravity would deflect starlight.

findings n.
研究或观测结果
the findings were hailed as confirming

hail as v.phr.
被盛赞为
the findings were hailed as confirming

eminent adj.
卓越的；著名的
Almost overnight, Einstein, eminent

rigorous adj.
严格缜密的
rigorous experimental confirmation,

confirmation n.
证实
rigorous experimental confirmation,

deem v.
认定；认为
deemed the empirical support for

empirical adj.
实证的；以观察实验为依据的
deemed the empirical support for

insufficient adj.
不足的
support for relativity insufficient.

criterion n.
标准
nomination fulfilled Nobel's criteria,

reserve v.
保留；暂缓颁发
the 1921 prize was reserved for a

confer on phr.v.
与on连用：授予
Swedish Academy of Sciences conferred

underpin v.
为……奠定基础
helped underpin quantum theory.

clarify v.
澄清；明确说明
letter the next day clarified that the

take no account of v.phr.
不予考虑
prize took no account of relativity.

bound for adj.phr.
驶往；开往
then aboard a steamer bound for Japan,

fundamental adj.
基本的；根本的
to the fundamental ideas and problems

honour v.
表彰；授予荣誉
by the man the prize had honoured.

文中词汇与搭配

补充词汇共33项。保留每个表达在本文中的意思。

observe v.
观测

general relativity n.phr.
广义相对论

bold adj.
大胆的

gravity n.
引力

joint adj.
联合的；联席的

proclaim v.
宣告

headline n.
新闻标题

overnight adv.
一夜之间

physicist n.
物理学家

celebrity n.
名人

dazzle v.
使赞叹；使目眩

experimental adj.
实验的

committee n.
委员会

specialist n.
专家

nomination n.
提名

fulfil v.
符合；满足

statute n.
章程；成文规定

permit v.
允许

on prep.
标示获授予的人

cite v.
列举；引作表彰理由

govern v.
支配；决定

eject v.
使喷出；释放出

electron n.
电子

quantum adj.
量子的

aboard prep.
在船上；乘船

steamer n.
轮船

ceremony n.
典礼

deliver v.
发表演讲

lecture n.
学术演讲；讲座

congress n.
大会

devote to phr.v.
与to连用：专门用于

to prep.
标示投入的对象

podium n.
讲台

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the sun's gravity would _____ starlight.

02 valued _____ experimental confirmation,

03 helped _____ quantum theory.

04 the prize _____ relativity.

文章主线

- 成名 1919年日食观测被誉为证实太阳引力会使星光偏折，这一相对论预言让爱因斯坦迅速举世闻名。
- 分歧 委员古尔斯特兰德认为相对论证据不足；1921年因无提名达标，物理学奖按章程保留一年。
- 授奖 1922年，保留的1921年度物理学奖授予爱因斯坦，表彰理论物理贡献，尤其是发现光电效应定律。
- 区别 光电效应指光使金属释放电子，这一定律帮助奠定量子理论基础；授奖决定次日的信件明确：该奖不考虑相对论。
- 演讲 爱因斯坦赴日错过颁奖典礼；1923年，他的诺贝尔演讲却专讲相对论，让被奖项绕开的理论登上讲台。

参考答案　1 deflect 2 rigorous 3 underpin 4 took no account of

末日种子库的第一次取种

The Day the Doomsday Vault Opened



全球种子保险箱第一次为谁打开?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

On Spitsbergen, in Norway's Svalbard archipelago, a corridor has been drilled deep into a frosted hillside. At its end, cartons of precious seed sit silently in a sub-zero repository, insulated by permanently frozen ground.

The Svalbard Global Seed Vault opened near Longyearbyen in February 2008 to insure crops such as beans, rice, and wheat against the horrors people dread, from atomic war to disease. It has ample room for about four and a half million samples. Essentially the planet's ultimate backup, it is run like a bank's safe-deposit box. Depositors keep ownership, and only they can take their seeds out.

The first retrieval came not after an asteroid strike but amid strife on the ground. ICARDA, a research centre whose gene bank stood near Aleppo, had moved its headquarters to Beirut during Syria's civil war. In September 2015 it asked for about 116,000 samples, 130 of the 325 boxes it had sent to the Arctic before the hostilities. They included robust strains of wheat and barley tolerant of drought and heat.

In Lebanon and Morocco, ICARDA sowed the seeds, and vigorous plants sprouted to regenerate its collection. Once multiplication had gone far enough, a homecoming began in 2017, consignment by consignment, restocking the Arctic stronghold.

That same year brought a scare. After record-breaking warmth, meltwater ran into the roughly 100-metre entrance tunnel and solidified. The intrusion stopped short of the seeds, which stayed safely at minus 18 degrees Celsius. Norway waterproofed the tunnel and dug drainage ditches.

A haven built against the apocalypse had a weakness at its own entrance, and warmer weather found it.

中文译文

在挪威斯瓦尔巴群岛的斯匹次卑尔根岛，一条通道深深钻入结霜的山坡。通道尽头，一箱箱珍贵种子静静存放在零度以下的储藏库里，永久冻土为它们提供隔热保护。

斯瓦尔巴全球种子库于2008年2月在朗伊尔城附近启用，为豆类、水稻、小麦等作物提供保障，以防人们畏惧的种种灾难，从核战争到病害。库内空间充足，可容纳约450万份样本。它实质上是地球最后一道种子备份，运作方式类似银行保险箱：存放者保留种子的所有权，也只有他们有权取回。

这座种子库的首次取种，并非发生在小行星撞击之后，而是在地面战乱之中。研究中心ICARDA的基因库原设在阿勒颇附近，叙利亚内战期间，其总部迁往贝鲁特。2015年9月，该中心申请取回约11.6万份样本，即战事发生前由它送往北极的325箱中的130箱。其中包括耐旱、耐热的健壮小麦和大麦品系。

ICARDA在黎巴嫩和摩洛哥播下这些种子，茁壮的植株萌发生长，重建了该中心自己的种子样本库。繁育数量足够多后，一场回归之旅于2017年开始：种子一批接一批送回，重新补充这座北极堡垒。

同年，种子库也经历了一场虚惊。创纪录的暖天气过后，融水流入约100米长的入口隧道，并凝结成冰。但水和冰没有抵达种子存放处，种子始终安全保存在零下18摄氏度。挪威随后给隧道加做防水，并挖出排水沟。

这座为抵御末日灾难而建的庇护所，弱点却在自己的入口，变暖的天气把它暴露了出来。

核心词汇

先记这22项。连着下方的原文短语一起回想。

corridor n.
通道；走廊
a corridor has been drilled

repository n.
储藏库
a sub-zero repository

insulate v.
隔热保护
insulated by permanently frozen ground

seed vault 名词短语
种子储藏库
The Svalbard Global Seed Vault

insure v.
为……提供保障
insure crops such as beans

ample adj.
充足的
ample room

backup n.
备份；后备
the planet's ultimate backup

ownership n.
所有权
Depositors keep ownership

retrieval n.
取回
The first retrieval

strife n.
冲突；战乱
amid strife on the ground

gene bank 名词短语
基因库；种质资源库
a research centre whose gene bank

headquarters n.
总部
moved its headquarters to Beirut

robust adj.
健壮的；适应力强的
robust strains of wheat and barley

strain n.
作物品系
robust strains of wheat and barley

tolerant of 形容词短语
能耐受……的
tolerant of drought and heat

sow v.
播种
ICARDA sowed the seeds

regenerate v.
重新繁育；恢复
to regenerate its collection

consignment n.
一批运送的货物
consignment by consignment

restock v.
重新补充库存
restocking the Arctic stronghold

intrusion n.
侵入
The intrusion stopped short

stop short of 动词短语
在到达……前停住
stopped short of the seeds

drainage n.
排水
dug drainage ditches

文中词汇与搭配

补充词汇共39项。保留每个表达在本文中的意思。

archipelago n.
群岛

drill v.
钻凿

frosted adj.
覆着霜的

hillside n.
山坡

carton n.
纸箱

precious adj.
珍贵的

silently adv.
静静地

sub-zero adj.
零度以下的

permanently adv.
永久地

bean n.
豆类

against prep.
防范；抵御

horror n.
可怕的灾难

dread v.
畏惧

atomic adj.
原子能的；核的

essentially adv.
实质上；基本上

ultimate adj.
最终的；最后一道的

safe-deposit box 名词短语
银行保险箱

depositor n.
存放者

asteroid n.
小行星

civil war 名词短语
内战

文中词汇与搭配 续

补充词汇共39项。保留每个表达在本文中的意思。

ask for 动词短语
申请取回；请求

Arctic 专有名词
北极地区

hostilities 复数名词
战事；敌对行动

barley n.
大麦

vigorous adj.
茁壮的；旺盛的

sprout v.
发芽

multiplication n.
繁殖；扩繁

homecoming n.
回归

stronghold n.
堡垒；坚固据点

scare n.
虚惊；惊慌

record-breaking adj.
破纪录的

tunnel n.
隧道

solidify v.
凝固

safely adv.
安全地

waterproof v.
做防水处理

ditch n.
沟渠

haven n.
庇护所；安全之地

apocalypse n.
末日灾难

weakness n.
弱点

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Depositors keep _____, and only they can take their seeds out.

02 The first _____ came not after an asteroid strike but amid strife on the ground.

03 robust strains of wheat and barley _____ drought and heat.

04 The _____ stopped short of the seeds, which stayed safely at minus 18 degrees Celsius.

文章主线

- 备份 2008年启用的斯瓦尔巴种子库，为防战争、病害保存农作物种子备份；种子归存放者，只有他们能取回。
- 取种 叙利亚内战中，农业研究中心ICARDA迁移总部；2015年取回自存的部分备份约11.6万份，这是种子库首次出库。
- 重建 该中心在黎巴嫩、摩洛哥繁育种子，恢复自己的种子样本库，并从2017年起分批将种子存回北极。
- 虚惊 同年，高温后的融水进入入口隧道并结冰，未触及种子；种子仍安全保存在零下18摄氏度。
- 加固 挪威给隧道加做防水并开挖排水沟，修补这座全球种子保险箱因天气变暖而暴露的入口弱点。

参考答案　1 ownership 2 retrieval 3 tolerant of 4 intrusion

郁金香狂热：泡沫与传说

Tulip Mania: The Bubble and the Legend



郁金香真的让整个荷兰破产了吗?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In the winter of 1636–37, Dutch merchants and brokers met in taverns to trade contracts for tulip bulbs still in the ground, a speculative business they called windhandel, "trading in the wind". Prices of prized "broken" tulips, with streaked petals, soared.

In early February 1637, at a sale in Haarlem, the hub of the trade, no one would bid. Days later, at an auction in Alkmaar, one bulb still fetched 5,200 guilders, the Dutch currency of the day, more than a decade's wages for a skilled craftsman. But the frenzy was over. Prices collapsed, many buyers defaulted, and unsettled contracts piled up. Why the market turned so swiftly is still contested.

Two centuries later, Charles Mackay, a Scottish journalist, recounted the episode in a book on popular delusions, casting it as national hysteria. In his telling, rich and poor gambled, great merchants were reduced almost to begging, and Dutch commerce reeled. His vivid account made tulip mania shorthand for bubbles fuelled by reckless buying.

Yet the archives tell a smaller story. Mackay's story relied largely on satirical pamphlets of 1637 that preached against greed. Anne Goldgar, a historian at King's College London, scoured Dutch records and counted only a few hundred traders, mostly merchants and craftsmen. She found no bankruptcy caused by tulips and no measurable harm to the economy. Many buyers were later allowed to settle their contracts by paying a small fraction of the price, and few were ruined.

The boom was real but fleeting. What spread, and still circulates nearly four centuries on, was the moral: a sermon drawn from those pamphlets and retold by Mackay as history.

中文译文

1636至1637年冬天，荷兰商人和经纪人聚在小酒馆里，交易还埋在土中的郁金香球茎合约。他们把这门投机生意称为windhandel，即“风中交易”。名贵的“碎色”郁金香花瓣带有条纹，价格一路飙升。

1637年2月初，在交易中心哈勒姆的一场买卖中，竟无人愿意出价。几天后，阿尔克马尔的一次拍卖仍有一只球茎售得5200荷兰盾——那是当时荷兰的货币，金额超过熟练工匠十年的工资。但狂热已经结束：价格暴跌，许多买家违约，未结清的合约越积越多。市场为何如此迅速转向，至今仍有争议。

两个世纪后，苏格兰记者查尔斯·麦凯在一本讨论群体性错觉的书中重述此事，将它描绘成全国性的狂热。在他的笔下，穷人富人都参与赌博，大商人几乎沦为乞丐，荷兰商业遭到重创。他生动的叙述使“郁金香狂热”成为不计后果的买入所催生的投机泡沫的代名词。

然而，档案呈现的规模要小得多。麦凯主要依据1637年那些讽刺贪婪、劝人戒贪的小册子来讲故事。伦敦国王学院历史学家安妮·戈德加仔细搜遍荷兰档案，只找到数百名交易者，大多是商人和工匠。她未找到因郁金香而破产的案例，也未发现经济遭受可衡量损害的证据。许多买家后来获准只支付原价的一小部分便结清合约，倾家荡产者寥寥无几。

这场热潮真实存在，却很短暂。真正广为传播、近四个世纪后仍在流传的，是其中的道德教训：一篇源于那些小册子的训诫，被麦凯当作历史重新讲述。

“未找到破产或经济损害的证据”是档案研究的发现，不等于断言无人蒙受损失。

核心词汇

先记这22项。连着下方的原文短语一起回想。

broker n.
经纪人；交易中介
merchants and brokers

contract n.
合约；合同
contracts for tulip bulbs

speculative adj.
投机性的
a speculative business

soar v.
飙升；猛涨
soared

bid v.
出价；竞价
no one would bid

fetch v.
售得；拍出某价格
one bulb still fetched 5,200 guilders

frenzy n.
狂热
the frenzy was over

collapse v.
暴跌；崩落
Prices collapsed

default v.
违约；未履行付款义务
many buyers defaulted

contest v.
受到争议或质疑
is still contested

hysteria n.
群体性狂热
national hysteria

shorthand for 名词短语
……的代名词
shorthand for bubbles

reckless adj.
不计后果的；鲁莽的
reckless buying

archive n.
档案；档案资料
the archives tell a smaller story

satirical adj.
讽刺性的
satirical pamphlets

pamphlet n.
小册子
satirical pamphlets

scour v.
彻底搜寻；仔细查遍
scoured Dutch records

bankruptcy n.
破产
no bankruptcy caused by tulips

measurable adj.
可以衡量的
no measurable harm to the economy

settle v.
结清；了结合约
settle their contracts

fraction n.
一小部分；分数
a small fraction of the price

fleeting adj.
短暂的；转瞬即逝的
real but fleeting

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

merchant n.
商人

tavern n.
小酒馆

tulip n.
郁金香

streaked adj.
带条纹的

petal n.
花瓣

hub n.
中心；枢纽

auction n.
拍卖会

currency n.
货币

wage n.
工资；工钱

unsettled adj.
未结清的

pile up 动词短语
堆积；越积越多

swiftly adv.
迅速地

journalist n.
记者

recount v.
讲述；叙述

episode n.
事件；一段往事

popular delusion 名词短语
群体性错觉；大众妄想

gamble v.
赌博

commerce n.
商业；贸易

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

reel v.
遭受冲击；摇摇欲坠

vivid adj.
生动的；鲜明的

mania n.
狂热

bubble n.
投机泡沫

rely v.
依赖；依据

on prep.
表示依赖或依据的对象

preach v.
宣扬；说教

against prep.
反对；抵制

greed n.
贪婪

historian n.
历史学家

harm n.
损害

economy n.
经济

a art.
不定冠词，表示一份或一部分

of prep.
……的

boom n.
繁荣；热潮

circulate v.
流传；传播

moral n.
寓意；道德教训

sermon n.
训诫；说教

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Prices collapsed, many buyers _____, and unsettled contracts piled up.

02 Why the market turned so swiftly is still _____.

03 Mackay's story relied largely on _____ pamphlets of 1637 that preached against greed.

04 She found no bankruptcy caused by tulips and no _____ harm to the economy.

文章主线

- 热潮 1636至1637年，荷兰商人和经纪人交易尚在土里的郁金香球茎合约，稀有品种价格暴涨。
- 破裂 1637年2月买盘骤冷，仍有球茎卖5200盾、抵工匠十余年收入；随后价格崩落、违约堆积，原因仍有争议。
- 传说 两个世纪后，麦凯主要依据讽刺贪婪的小册子，把它写成全民投机、商人破产、经济重创的故事。
- 档案 戈德加查档只找到数百名交易者，未找到郁金香引发破产或经济遭受可衡量损害的证据。
- 辨别 许多买家只付原价一小部分便结清合约，鲜有人倾家荡产；热潮真实，广传的全国灾难故事却被放大。

参考答案　1 defaulted 2 contested 3 satirical 4 measurable

一座火山，造出了《弗兰肯斯坦》

The Volcano That Gave Us Frankenstein



远在热带的火山，怎样与《弗兰肯斯坦》相连?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In June 1816 a ceaseless downpour kept a party besieged at the Villa Diodati, a mansion the poet Lord Byron had leased on the outskirts of Geneva. After reading an anthology of German ghost stories in French translation, Byron proposed that each guest write a supernatural tale.

The melancholy weather was attributable to a distant blast. On 10 April 1815 Mount Tambora, on the Indonesian island of Sumbawa, blew itself apart in one of the most potent eruptions in recorded history. About 10,000 people died immediately, and tens of thousands more of starvation and disease in the aftermath.

The volcano pumped prodigious quantities of sulphur dioxide from its crater into the stratosphere, where it formed sulphate aerosols that blanketed the globe, reflecting sunshine and thereby lowering temperatures world-wide. Europe and North America came to know 1816 as the Year Without a Summer.

Europe's summer was chilly and relentlessly wet. Crops failed or lay rotten in saturated fields, cereal and bread prices escalated, and scarcity, disease, and unrest followed.

The eighteen-year-old Mary Godwin, later Mary Shelley, had set out in May alongside Percy Bysshe Shelley and Claire Clairmont. In her 1831 preface she would recollect a damp, gloomy summer whose torrents often kept the group indoors for days.

Byron's challenge proved fruitful. His physician John Polidori produced The Vampyre, whilst Byron's poetic output included Darkness. Mary began the ghastly yarn that became Frankenstein, which appeared anonymously in 1818 and was later reissued under her name. It is often called the founding work of the science-fiction genre.

A tropical volcano had supplied the rain, and a teenager the monster.

中文译文

1816年6月，连绵的倾盆大雨把一群人困在迪奥达蒂别墅。这座宅邸位于日内瓦郊外，由诗人拜伦勋爵租住。读过一本德国鬼故事选集的法文译本后，拜伦提议每位客人各写一个超自然故事。

这阴郁的天气，源于遥远的一次猛烈喷发。1815年4月10日，印度尼西亚松巴哇岛上的坦博拉火山炸裂，成为有记录以来最猛烈的火山喷发之一。约一万人当场死亡，随后又有数万人死于饥饿和疾病。

火山把大量二氧化硫从火山口送进平流层。这些气体在那里形成硫酸盐气溶胶，覆盖全球、反射阳光，从而使世界各地降温。欧洲和北美后来将1816年称为“无夏之年”。

欧洲的夏天寒冷而持续多雨。庄稼不是歉收，就是烂在浸透水的田里；谷物和面包价格飞涨，物资短缺、疾病与社会动荡接踵而来。

18岁的玛丽·戈德温，也就是后来的玛丽·雪莱，早在5月便与珀西·比希·雪莱、克莱尔·克莱蒙特一同启程。她在1831年的序言中回忆，那是一个潮湿阴沉的夏天，大雨常让一行人接连几天无法出门。

拜伦的提议结出了丰硕成果。他的医生约翰·波利多里写出《吸血鬼》，拜伦创作的诗歌中则有《黑暗》。玛丽开始写那个恐怖故事，它后来成为《弗兰肯斯坦》：1818年匿名出版，之后以她的名字再版。这部作品常被称为科幻文学的开山之作。

热带的一座火山带来了大雨，一位少女则创造了怪物。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ceaseless adj.
无休止的；持续不断的
a ceaseless downpour

besiege v.
围困；使无法离开
kept a party besieged

anthology n.
选集
an anthology of German ghost stories

attributable to 形容词短语
可归因于
attributable to a distant blast

potent adj.
强大的；猛烈的
one of the most potent eruptions

aftermath n.
灾难之后；后果
in the aftermath

prodigious adj.
数量庞大的
prodigious quantities

sulphur dioxide 名词短语
二氧化硫
quantities of sulphur dioxide

aerosol n.
气溶胶；悬浮的微小颗粒
sulphate aerosols

thereby adv.
从而
thereby lowering temperatures

relentlessly adv.
持续不断地
chilly and relentlessly wet

saturated adj.
浸透水的
saturated fields

escalate v.
迅速上涨
bread prices escalated

scarcity n.
短缺
scarcity, disease, and unrest

unrest n.
社会动荡
scarcity, disease, and unrest

preface n.
序言
In her 1831 preface

recollect v.
回忆
she would recollect

fruitful adj.
成果丰硕的
Byron's challenge proved fruitful

anonymously adv.
匿名地
appeared anonymously in 1818

reissue v.
重新出版
reissued under her name

founding work 名词短语
开山之作
the founding work

genre n.
文学体裁；类型
the science-fiction genre

文中词汇与搭配

补充词汇共44项。保留每个表达在本文中的意思。

downpour n.
倾盆大雨

mansion n.
大宅；宅邸

poet n.
诗人

lease v.
租用

outskirts n.
郊外

ghost story 名词短语
鬼故事

translation n.
译本

supernatural adj.
超自然的

melancholy adj.
阴郁的

Indonesian adj.
印度尼西亚的

blow v.
爆炸；与apart连用指炸裂

apart adv.
成碎片；与blow连用指炸裂

eruption n.
火山喷发

tens of thousands 数量短语
数万

starvation n.
饥饿；饿死

volcano n.
火山

pump v.
大量送入；喷出

crater n.
火山口

blanket v.
覆盖；笼罩

sunshine n.
阳光

lower v.
降低

world-wide adv.
在全世界范围内

文中词汇与搭配 续

补充词汇共44项。保留每个表达在本文中的意思。

know v.
把……称为；与as连用

as prep.
作为；引出称呼

chilly adj.
寒冷的

crop n.
庄稼

fail v.
歉收；收成不好

rotten adj.
腐烂的

cereal n.
谷物

eighteen-year-old adj.
十八岁的

set out 动词短语
启程

alongside prep.
与……一起

damp adj.
潮湿的

gloomy adj.
阴沉的

torrent n.
倾盆大雨

whilst conj.
而；与此同时

poetic adj.
诗歌的

output n.
创作成果

ghastly adj.
恐怖的

yarn n.
故事

Frankenstein 专名
《弗兰肯斯坦》

science fiction n.（作定语）
科幻文学

tropical volcano 名词短语
热带火山

monster n.
怪物

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The melancholy weather was _____ a distant blast.

02 Byron's challenge proved _____.

03 which appeared _____ in 1818 and was later reissued under her name.

04 It is often called the founding work of the science-fiction _____.

文章主线

- 喷发 1815年，印尼坦博拉火山喷发，当场约一万人死亡，随后又有数万人死于饥饿和疾病。
- 降温 火山喷出的二氧化硫在高空形成硫酸盐微粒，反射阳光降温，使1816年成为欧美无夏之年。
- 灾情 欧洲夏季寒冷多雨，庄稼歉收、粮价上涨，饥饿、疾病与社会动荡接踵而来。
- 契机 大雨把拜伦、玛丽·雪莱等人困在日内瓦郊外的别墅，拜伦提议每人写一篇恐怖故事。
- 成果 18岁的玛丽开始写《弗兰肯斯坦》，1818年匿名出版，这部作品后来常被视为科幻小说的开山之作。

参考答案　1 attributable to 2 fruitful 3 anonymously 4 genre

从车库到世界之巅

From a Garage to the Top of the World



离开苹果后，乔布斯怎样走回苹果？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In May 1985, after a bruising confrontation, Apple's board ultimately backed chief executive John Sculley. Steve Jobs, 30, was stripped of his managerial duties at the company he had co-founded. That September, sidelined and alienated, he resigned.

Rewind three decades. Born in San Francisco on 24 February 1955, Jobs was adopted as a newborn. On 1 April 1976 he founded Apple with Steve Wozniak, the electronics engineer behind its first prototypes. Apple's garage origins are largely symbolic; Wozniak calls their significance inflated. Apple floated on 12 December 1980; the flagship Macintosh debuted on 24 January 1984.

Exile proved productive. Later in 1985 Jobs founded NeXT, a computer start-up. In February 1986 he acquired Lucasfilm's computer-graphics outfit, which became Pixar. It began as a manufacturer of specialised hardware, not chiefly an animation studio. Toy Story, released on 22 November 1995, was the first feature-length computer-animated film.

The reversal came on 20 December 1996: Apple agreed to buy NeXT, gaining the core software of the eventual Mac OS X and reuniting with its estranged co-founder. In August 1997 Microsoft invested $150 million in its distressed competitor; that September Jobs took the reins as interim chief. Apple, he later said, had been weeks from ruin. The iMac, iPod, and iPhone followed, the iPhone showcased on 9 January 2007.

In October 2003 Jobs received a diagnosis: an uncommon, cancerous pancreatic tumour. In August 2011 Apple briefly overtook Exxon Mobil as the world's most valuable company. His tenure ended on 24 August 2011; he died on 5 October 2011, aged 56. He had spent nearly fourteen years at the helm of the company that ousted him.

中文译文

1985年5月，经历一场激烈交锋后，苹果董事会最终支持了首席执行官约翰·斯卡利。30岁的史蒂夫·乔布斯，在自己参与创办的公司里被解除管理职务。同年9月，遭到架空和疏远的他辞职了。

把时间倒回三十年前。乔布斯于1955年2月24日生于旧金山，出生不久便被收养。1976年4月1日，他和史蒂夫·沃兹尼亚克创办苹果；首批样机由电子工程师沃兹尼亚克研制。苹果的“车库起源”很大程度上具有象征意义，沃兹尼亚克认为它的重要性被夸大了。苹果于1980年12月12日上市，旗舰产品Macintosh于1984年1月24日首次推出。

离开苹果后的岁月颇有成果。1985年晚些时候，乔布斯创办电脑初创公司NeXT。1986年2月，他收购卢卡斯影业的电脑图形团队，后来发展为皮克斯。皮克斯起初制造专用硬件，主要还不是动画工作室。1995年11月22日上映的《玩具总动员》，成为首部长篇电脑动画电影。

转折发生在1996年12月20日：苹果同意收购NeXT，获得后来Mac OS X所用的核心软件，也让苹果与关系疏远的联合创始人乔布斯重新走到一起。1997年8月，微软向陷入困境的竞争对手苹果投资1.5亿美元；同年9月，乔布斯接掌大权，出任临时负责人。他后来回忆，当时苹果离破产只差几周。此后，iMac、iPod和iPhone相继问世，其中iPhone于2007年1月9日公开亮相。

2003年10月，乔布斯被诊断患有一种罕见的恶性胰腺肿瘤。2011年8月，苹果一度超过埃克森美孚，成为全球市值最高的公司。乔布斯于2011年8月24日卸任，10月5日去世，终年56岁。在那家曾经将他逐出的公司里，他重新掌舵了将近十四年。

eventual Mac OS X指后来出现的系统，不是某个“最终版本”；微软投资的对象是苹果。

核心词汇

先记这22项。连着下方的原文短语一起回想。

confrontation n.
对抗；交锋
a bruising confrontation

board n.
董事会
Apple's board

executive n.
高管；执行官
chief executive John Sculley

be stripped of 动词短语
被剥夺
was stripped of his managerial duties

managerial adj.
管理的
his managerial duties

sideline v.
使失去实权；架空
sidelined and alienated

resign v.
辞职
he resigned

prototype n.
样机
its first prototypes

symbolic adj.
象征性的
largely symbolic

significance n.
重要性；意义
their significance inflated

productive adj.
富有成果的
Exile proved productive

acquire v.
收购
acquired Lucasfilm's

manufacturer n.
制造商
a manufacturer of specialised hardware

reversal n.
局面逆转
The reversal came

eventual adj.
后来出现的；最终的
the eventual Mac OS X

estranged adj.
关系疏远的
its estranged co-founder

distressed adj.
陷入财务困境的
its distressed competitor

competitor n.
竞争对手
its distressed competitor

interim adj.
临时的
as interim chief

diagnosis n.
诊断
received a diagnosis

tenure n.
任期
His tenure ended

at the helm of 介词短语
掌管；领导
at the helm of the company

文中词汇与搭配

补充词汇共42项。保留每个表达在本文中的意思。

bruising adj.
激烈而令人受挫的

ultimately adv.
最终

co-found v.
共同创办

alienate v.
使疏远

rewind v.
倒回；回溯

San Francisco n.
旧金山

adopt v.
收养

newborn n.
新生儿

found v.
创办

electronics n.
电子技术

origin n.
起源

inflate v.
夸大

float v.
上市；公开发行股票

flagship adj.
旗舰的；主打的

debut v.
首次推出

exile n.
被迫离开的时期

start-up n.
初创公司

graphics n.
电脑图形

outfit n.
团队；机构

Pixar n.
皮克斯

specialised adj.
专用的

文中词汇与搭配 续

补充词汇共42项。保留每个表达在本文中的意思。

hardware n.
硬件

animation n.
动画

studio n.
工作室

release v.
上映；发行

feature-length adj.
长篇电影长度的

computer-animated adj.
电脑动画制作的

core adj.
核心的

reunite with 动词短语
与……重聚

invest v.
投资

in prep.
向；对（投资对象）

take the reins 动词短语
接掌大权

ruin n.
破产；毁灭

showcase v.
公开展示

uncommon adj.
罕见的

cancerous adj.
恶性的

pancreatic adj.
胰腺的

tumour n.
肿瘤

briefly adv.
短暂地

overtake v.
超过

valuable adj.
最有价值的；此处指市值最高

oust v.
逐出

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In May 1985, after a bruising _____, Apple's board ultimately backed chief executive John Sculley.

02 On 1 April 1976 he founded Apple with Steve Wozniak, the electronics engineer behind its first _____.

03 In February 1986 he _____ Lucasfilm's computer-graphics outfit, which became Pixar.

04 In August 1997 Microsoft invested $150 million in its distressed competitor; that September Jobs took the reins as _____ chief.

文章主线

- 创业 1976年，乔布斯与沃兹尼亚克创办苹果，后者研制首批样机；苹果1980年上市，车库起源的意义被夸大。
- 离开 1985年，董事会在冲突中支持首席执行官斯卡利，解除乔布斯管理职务；被架空的乔布斯随后辞职。
- 另起 他创办NeXT，收购皮克斯前身；皮克斯原造专用硬件，1995年推出全球首部长篇电脑动画《玩具总动员》。
- 回归 1996年苹果同意收购NeXT，获得后来Mac OS X的核心软件；次年微软向苹果注资1.5亿美元，乔布斯回归掌舵。
- 重振 iMac、iPod、iPhone相继推出，苹果2011年一度市值全球第一；患胰腺恶性肿瘤的乔布斯同年卸任、去世。

参考答案　1 confrontation 2 prototypes 3 acquired 4 interim